Proposte di Tesi – Prof. Martinenghi

1. Enhancing RAG through Advanced Ranking

Priorità: ALTA — Sfrutta direttamente la tua esperienza RAG + pgvector. Deliverable chiaro: prototipo pipeline + benchmark.

- **Collaboration** with Prof. Paolo Papotti (EURECOM, France)

Background.

- Ranking: scoring functions and user preferences.
- RAG: retrieving, ranking, and feeding contextual knowledge into LLMs via embeddings.
- Ranking (and re-ranking) scoring functions in RAG (Cosine Similarity, Dot Product, Euclidean Distance, BM25, etc.) determine which pieces of evidence are actually seen by the model, and in which order.

Project Objectives.

- Explore ranking techniques using uncertain scoring functions to optimize RAG.
- Assess the impact on quality and efficiency of LLM outputs.

Example of thesis outcome.

- Prototype RAG pipelines with advanced ranking/re-ranking modules.
- Systematic evaluation of the integration of ranking methodologies in RAG.
- Guidelines and enhanced/revised performance metrics definition for LLMs in context-rich environments.

Key references.

- Lewis, P. et al., *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*, NeurIPS 2020.
- Karpukhin, V. et al., *Dense Passage Retrieval for Open-Domain Question Answering*, ACL 2020.

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2. LLMs & DBs

Priorità: ALTA — Vicina alla tua esperienza RAG + PostgreSQL. Esplorativa, con libertà di scope.

- **Collaboration** with Prof. Paolo Papotti (EURECOM, France)

Background.

- RAG: embedding, retrieving, and feeding context into LLMs.

Project Objectives.

- Explore innovative query answering tools beyond standard SQL capabilities.
- Experiment with a variety of LLMs, datasets, and queries.

Example of thesis outcome.

- New insights into the extended possibilities of query answering systems.

Key references.

- Piantanida, S., *Evaluation of LLMs Capabilities for Table Question Answering on Incomplete Structured Data*, Master's Thesis, Politecnico di Milano, 2025.

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3. Fairness & LLM Guardrailing

Priorità: ALTA — Collaborazione con Banca d'Italia (ottimo per CV). Python, LLM, tema attualissimo.

- **Collaboration** with the Bank of Italy

Background.

- Data Mining or Machine Learning basics.
- Python language.

Project Objectives.

- Elaborate a solution that handles Fairness in LLMs through Guardrailing (i.e., ensuring that the generated content is safe and adheres to specified guidelines).

Example of thesis outcome.

- A thorough audit of fairness methods in selected, open-source LLMs.
- A formalized methodology for adapting fairness metrics from classification to generative text guardrailing.
- The implementation and evaluation of a novel guardrail model.

Key references.

- https://dl.acm.org/doi/abs/10.1145/3457607
- https://arxiv.org/abs/2112.04359

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4. Fairness & Data Quality in ML Classification

Priorità: MEDIA — Scope ben definito, pipeline esistente da migliorare. Rischio basso, ma meno stimolante.

Background.

- Data Mining or Machine Learning basics.
- Python language.

Project Objectives.

- Evaluate and improve a solution that handles Fairness and Data Quality based on Approximate Conditional Functional Dependencies (ACFDs).

Example of thesis outcome.

- Design and implement an enhanced fairness-repair phase for the existing data-preparation pipeline, reducing reliance on deletion-based correction.
- Demonstrate that the new repair strategy improves both fairness metrics and classification performance on multiple ML models.

Key references.

- https://dl.acm.org/doi/abs/10.1145/3457607
- https://ceur-ws.org/Vol-3194/paper66.pdf
- https://www.iris.unisa.it/retrieve/e2915b31-af22-8981-e053-6605fe0a83a3/TKDE-published.pdf
- Alireza Tahmasebi, *FairDataFlow: a Modern Solution to Ensure Data Quality and Fairness*, Master's Thesis, Politecnico di Milano, 2025.

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5. Fairness Intersectionality in ML Classification

Priorità: MEDIA — Python, ML, demo da produrre. Collaborazione con U. of Michigan. Meno connesso al tuo background LLM/RAG.

- **Collaboration** with Prof. H V Jagadish (University of Michigan, USA)

Background.

- Data Mining or Machine Learning basics.
- Python language.

Project Objectives.

- Formulate a new approach to solve fairness intersectionality issues in machine learning classification.

Example of thesis outcome.

- Find heuristics to find discriminated subgroups.
- Elaborate a mitigation strategy to solve fairness issues for discriminated subgroups.
- Produce a demo/code to perform such tasks.

Key references.

- https://dl.acm.org/doi/pdf/10.1145/3588433
- https://arxiv.org/pdf/2305.06969
- https://par.nsf.gov/servlets/purl/10535250

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6. Post-hoc Explainability in RAG: Impact on Fairness

Priorità: BASSA — Tema interessante ma scope vago e molto research-oriented. Rischio di perdersi.

- **Collaboration** with Prof. Kostas Stefanidis (Tampere University, Finland)

Background.
- RAG explainability is often post-hoc, relying on input perturbation or counterfactual retrieval.
- These interventions can change retrieval/generation behavior, potentially shifting both utility and fairness.

Project Objectives.
- Quantify utility/fairness trade-offs.
- Test fairness-aware re-ranking strategies.

Example of thesis outcome.
- Evaluation protocol + benchmark definition; empirical evidence on when explainability helps/hurts fairness.
- Prototype dashboard and guidelines definition for "fairness-aware" explanation settings.

Key references.
- Rorseth, J. et al., *RAGE Against the Machine: Retrieval-Augmented LLM Explanations.*
- Zhao, Y. et al., *Fairness and explainability: bridging the gap towards fair model explanations.*

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7. ER & SQL

Priorità: BASSA — Implementativo ma non AI/data science. Tool di traduzione ER->SQL, lontano dai tuoi interessi.
- **Collaboration** with Prof. Paolo Ciaccia (University of Bologna, Italy)

Background.
- E/R diagrams capture rich semantic constraints (cardinalities, hierarchies, weak entities, etc.).
- The relational schema only enforces a subset of the constraints, while others remain implicit.
- Automatically generating SQL constraints and triggers from E/R features can make schemas more faithful to the conceptual design and easier to maintain.

Project Objectives.
- Formalise how main E/R elements translate into additional SQL constraints and triggers, and design a configurable framework that takes an E/R model as input and automatically generates them.
- Study trade-offs between faithfulness to the conceptual model, readability of the generated schema, and runtime overhead of triggers.

Examples of thesis outcomes.
- Catalogue of E/R features and corresponding SQL constraint/trigger patterns.
- Prototype tool that parses an E/R specification and generates executable SQL schemas with embedded constraints and triggers.
- Development of a translation tool and evaluation on realistic case studies.

Key references.
- Francesco Moretti, *Traduzione automatica di schemi E/R: relazioni e vincoli*, Tesi di Laurea in Ingegneria Informatica, Università di Bologna.
- Chenming Zhao, *Translation of E/R Diagrams to SQL Code with Constraints and Triggers: Collapse of Hierarchies*, Master's Thesis, Politecnico di Milano, 2025.

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8. Linear top-k queries

Priorità: BASSA — Teoria pura: dimostrazioni, complessità computazionale, gap teorici. L'opposto di quello che cerchi.

- **Collaboration** with Prof. Paolo Ciaccia (University of Bologna, Italy)

Background.

- Linear top-k queries rank tuples via a linear scoring function.
- The skyline contains all Pareto-optimal points (i.e., those that are top-1 for some monotonic scoring function, not necessarily a linear one).
- Open question: how well can linear top-k queries cover the skyline, and how hard is it to retrieve all optimal points?

Project Objectives.

- Compute, for a given k, the minimum number q of linear top-k queries needed to retrieve the skyline.
- Compute, for a given q, the smallest k for which q linear top-k queries are able to retrieve the skyline.
- Study how to compute these indicators beyond trivial cases and their application as dataset descriptors.

Examples of thesis outcomes.

- Close theory gaps for simpler cases.
- Devise a general algorithm and evaluate its complexity.

Key references.

- Ciaccia & Martinenghi, *Directional queries: Making top-k queries more effective in discovering relevant results*, Proc. ACM Management of Data, 2024.
- Marco Somaschini, *Measuring and enhancing linear top-k queries' ability to discover the skyline*, Master's Thesis, Politecnico di Milano, 2022.
- Andrea Arbasino, Stefano Fumagalli, *Directional queries: a proposal for enhancing top-k queries*, Master's Thesis, Politecnico di Milano, 2023.

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9. Adding Diversity in Directional Query

Priorità: BASSA — Definizioni formali, algoritmi esatti, branch-and-bound. Molto accademico/teorico.

- **Collaboration** with Prof. H V Jagadish (University of Michigan, USA)

Background. Directional queries and diversified ranking in high-dimensional spaces.

Project Objectives.

- Identify application scenarios where directional preferences and diversity requirements naturally arise.
- Study the trade-offs between directional relevance and diversity in ranked result sets.
- Explore modeling choices and algorithmic strategies for combining directional and diversity criteria in a unified query framework.

Examples of thesis outcomes.

- Formal definition and analysis of directional-diversity problems and their properties.
- Design and implementation of exact and heuristic algorithms (e.g., improved greedy, branch-and-bound, "diverse neighbors" variants).
- Experimental evaluation on synthetic/real datasets: trade-offs between proximity to the query, attribute balance, and diversity.

Key references.

- Ciaccia and Martinenghi, *Directional queries: Making top-k queries more effective in discovering relevant results*, Proc. ACM Management of Data, 2024.
- Guo et al., *Finding diverse neighbors in high dimensional space*, ICDE 2018.

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10. UF-Skyline

Priorità: BASSA — Stessa famiglia teorica delle directional queries. Formalizzazione e algoritmi, poco implementativo.

- **Collaboration** with Prof. Paolo Ciaccia (University of Bologna, Italy)

Background.

- Top-k queries based only on utility can strongly violate fairness (i.e., unfair group representation).
- The UF-skyline returns all k-sets for which no other set has both higher utility and better fairness, thus exposing a full spectrum of utility/fairness trade-offs instead of a single solution.

Project Objectives.

- Formalise the UF-skyline framework for fair ranking in different settings.
- Understand how different fairness definitions affect the UF frontier and design algorithmic principles to enumerate UF-skyline k-sets efficiently, avoiding unnecessary branching and duplicates.
- Explore how UF-skyline results can support the choice among alternative utility/fairness trade-offs.

Examples of thesis outcomes.

- Implementation and experimental evaluation of UF-skyline algorithms handling different parameter configurations.

Key references.

- Liu et al., *Fair Top-k Query on Alpha-Fairness*, ICDE 2024.
- Asudeh et al., *Designing Fair Ranking Schemes*, SIGMOD 2019.

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